EVALUATING LIMITS

Rationalizing with the conjugate

Multiplying by the conjugate over itself changes the form, not the value: the radical leaves the numerator and the factor behind 0/0 comes into the open.

01

SUBSTITUTION GIVES 0/0 — BEFORE

$$\lim_{x \to 0} \frac{\sqrt{x+1} - 1}{x}$$

↓ `Multiply by the conjugate` — numerator and denominator by √(x + 1) + 1

02

CONJUGATE IN

$$\lim_{x \to 0} \frac{(\sqrt{x+1} - 1)(\sqrt{x+1} + 1)}{x(\sqrt{x+1} + 1)}$$

↓ `Difference of squares on top` — (x + 1) − 1 = x

03

FACTOR X EXPOSED

$$\lim_{x \to 0} \frac{x}{x(\sqrt{x+1} + 1)}$$

↓ `Cancel x` — the factor that caused the 0/0

04

SUBSTITUTE

AFTER

$$\lim_{x \to 0} \frac{1}{\sqrt{x+1}+1} = \frac{1}{2}$$

TOP AND BOTTOM, BOTH